

Create a Spring Web Project using Maven

SOLUTION:

1) Initial Setup:



Project

☐ Gradle - Groovy ☐ Gradle - Kotlin ☒ **Java** ☐ Kotlin ☐ Groovy

☒ **Maven**

Spring Boot

☐ 4.1.1 (SNAPSHOT) ☒ **4.1.0** ☐ 4.0.8 (SNAPSHOT) ☐ 4.0.7 ☐ 3.5.16

Project Metadata

Group

Artifact

Package name

Packaging ☒ **Jar** ☐ War

Configuration ☒ **Properties** ☐ YAML

Java ☐ 26 ☐ 25 ☐ 21 ☒ **17**

Dependencies

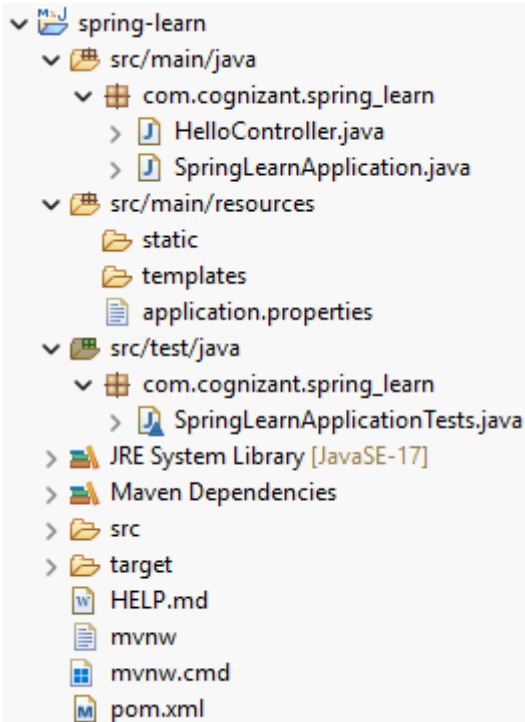
ADD DEPENDENCIES... CTRL + B

Spring Web **WEB**

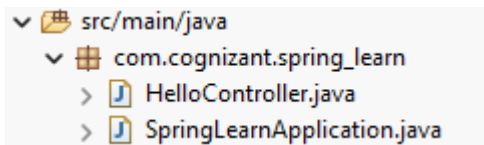
Build web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container.

Spring Boot DevTools **DEVELOPER TOOLS**

Provides fast application restarts, LiveReload, and configurations for enhanced development experience.



2) src/main/java:



HelloController.java

```
package com.cognizant.spring_learn;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;
@RestController
public class HelloController {
    @GetMapping("/")
    public String hello() {
        return "Spring Learn Application Running Successfully";
    }
}
```

SpringLearnApplication.java

```
package com.cognizant.spring_learn;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class SpringLearnApplication {

    public static void main(String[] args) {

        System.out.println("Inside Main Method");

        SpringApplication.run(
            SpringLearnApplication.class,
            args);

        System.out.println("Application Started");
    }
}
```

Purpose of @SpringBootApplication

@SpringBootApplication

Combination of three annotations:

@Configuration

@Configuration

Used to define Spring Beans.

@EnableAutoConfiguration

@EnableAutoConfiguration

Automatically configures:

- Tomcat
- Spring MVC
- Jackson
- Logging

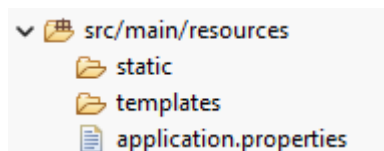
@ComponentScan

@ComponentScan

Scans current package and sub-packages for:

@Controller
@Service
@Repository
@Component

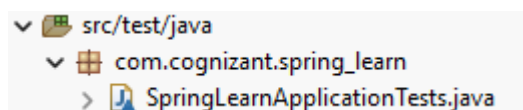
3) src/main/resources:



application.properties:

spring.application.name=spring-learn
server.port=8080

4) src/test/java:



SpringLearnApplicationTests.java

package com.cognizant.spring_learn;

import org.junit.jupiter.api.Test;
import org.springframework.boot.test.context.SpringBootTest;

@SpringBootTest

```
class SpringLearnApplicationTests {  
    @Test  
    void contextLoads() {  
    }  
}
```

5) pom.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    https://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>3.5.3</version>
    <relativePath/>
  </parent>

  <groupId>com.cognizant</groupId>
  <artifactId>spring-learn</artifactId>
  <version>0.0.1-SNAPSHOT</version>
  <name>spring-learn</name>

  <properties>
    <java.version>17</java.version>
  </properties>

  <dependencies>

    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-web</artifactId>
    </dependency>

    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-devtools</artifactId>
      <scope>runtime</scope>
      <optional>true</optional>
    </dependency>

    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-test</artifactId>
      <scope>test</scope>
    </dependency>

  </dependencies>

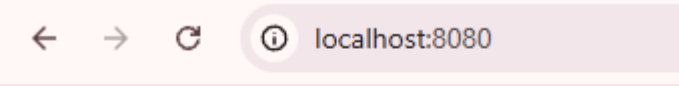
  <build>
    <plugins>
      <plugin>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-maven-plugin</artifactId>
      </plugin>
    </plugins>
  </build>
</project>
```

OUTPUT:

```
SpringLearnApplication [Java Application] C:\Users\Asus\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.9.v20251105-0741\jre\bin\javaw.exe (05-Jul-2026, 10:33:01 am elapsed 1:12:52) [pid: 22684]
Inside Main Method
Inside Main Method

:: Spring Boot ::                (v4.1.0)

2026-07-05T10:33:04.855+05:30 INFO 22684 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 21.0.9 with PID 22684 (C:\Users\Asus\
2026-07-05T10:33:04.859+05:30 INFO 22684 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-05T10:33:04.921+05:30 INFO 22684 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false'
2026-07-05T10:33:05.845+05:30 INFO 22684 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property
2026-07-05T10:33:05.864+05:30 INFO 22684 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-07-05T10:33:05.864+05:30 INFO 22684 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-05T10:33:05.864+05:30 INFO 22684 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-05T10:33:05.906+05:30 INFO 22684 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 984 ms
2026-07-05T10:33:06.329+05:30 INFO 22684 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-07-05T10:33:06.337+05:30 INFO 22684 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 1.916 seconds (process running for 3.956)
Application Started
2026-07-05T10:33:40.258+05:30 INFO 22684 --- [spring-learn] [nio-8080-exec-2] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-05T10:33:40.259+05:30 INFO 22684 --- [spring-learn] [nio-8080-exec-2] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-05T10:33:40.262+05:30 INFO 22684 --- [spring-learn] [nio-8080-exec-2] o.s.web.servlet.DispatcherServlet : Completed initialization in 3 ms
```



Spring Learn Application Running Successfully